Covariance

Hello, and welcome to another session on Business Statistics for Entrepreneurs. In the last module, we discussed confidence intervals and hypothesis testing. In this module, we are going to continue our journey on statistical inference and look at simple linear regression. Joining me in this journey are our grad students, Tejas and Ashwini, and along with them, I will cover some important concepts and relevant applications.

As always, you are going to keep me honest and on track. Welcome back Tejas and Ashwini. Are you ready for linear regression? Statistical inference was interesting. I realized rejecting the null hypothesis is a good thing. Yes, I never thought rejection would be so desirable. Yes, rejecting the null hypothesis indicates that the data provides conclusive evidence.

Okay, in this module, with linear regression, you will see confidence intervals and hypothesis tests come together to do more interesting things. So what are we going to cover? Let me start with an overview of what we are going to be covering in this module. We're going to start with two important concepts, namely covariance and correlation.

These two concepts explain the relationship between two variables obtained in a dataset. In particular, we will focus on linear relationship between two variables. In the process of going through these concepts, we will learn how to differentiate between causal and association relationship between various variables.

Then we will learn about simple linear regression that tries to establish a linear relationship between an independent variable x and a dependent variable y. Establishing such a relationship can be very powerful in predictive analytics. We will start by looking at how we can build a simple linear regression model.

In particular, we will look at the various stages involved in building a linear regression model. We will then understand the various assumptions behind a linear regression model. And then we will use the least square method to estimate the regression parameters for the model. After that, we will learn how to interpret the regression parameters under different functional forms.

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And then how to carry out model diagnostics to validate the model and seek improvements to the model where possible. Of course, we will blend all of this with applications of simple linear regression in several applications. Let us begin. Correlation and covariance are descriptive measures that explain the relationship between two variables.

To understand the concept of covariance and correlation, let us look at an example that we saw earlier in module one, when we looked at various types of datasets. If you recall, Hanumantha Pai working for Canara Bank in Mysore had collected data for monthly credit card expenditures of his customers. Do you remember what it had?

Did we not see this in the last module too? Perhaps with sampling distributions we did. If you recall, his random sample had data for 55 customers. Yes, he had recorded, one, the monthly amount charged in rupees to their credit card, secondly, the annual income in lakhs, and thirdly, the size of their household.

Yes, that is right. Now, when we computed the descriptive statistics, we had found that the average monthly charge on credit cards was 3852 rupees with a median of 3, 864 rupees and the standard deviation of this monthly charge was 1, 454 rupees. Now suppose Hanumantha wants to determine the relationship between the amount charged on the credit card each month versus the annual income of a household or the size of the household.

How many data points do we have? We have the 55 sample data points. Yes. And if we plot a scatter diagram between the amount of money charged on the y axis and the household income on the x axis, we see a positive relationship indicating that the higher the household income, higher is the associated monthly expense on the credit card.

The scatter diagram suggests that a straight line could be used as an approximation of this relationship. Yes, exactly. And next, if you look at a scatter plot between the amount of money charged on the y axis and the size of the household on the x axis. Again, we see a positive relationship, implying that higher the size of the household, higher is the monthly spend on the credit card.

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Yes, here too, a scatter diagram suggests that a straight line could be used as an approximation for this relationship. We will see next how covariance can be used as a descriptive measure of the linear association between two variables, such as the monthly spend on a credit card and the household income, or the monthly spend on a credit card and the size of a household.

Let us start by looking at covariance for a sample data that has n observations of two variables that we will label as x and y , where y could correspond to the monthly spend on the credit card, and X could correspond to the annual income of the household or the size of the household. Let us label our observation pairs as x one, y one, x two, y two, and so on all the way up to xnyn, where yi corresponds to the value of Y,

when X takes a value of xi. Let x bar be the average of the x values and y bar be the average of the y values. Then the sample covariance S of x y is defined as the sum of the products of xi minus x bar times yi minus y bar summed over all the n data pairs and then divided by n minus 1. Why divide n by 1?

The reason to divide by n minus 1 instead of n is the same degree of freedom argument that we used when we computed the sample variance in module 1. Note that in this formula we pair every xi, for example household income, with a yi which would be monthly spent and then we sum the products obtained by multiplying the deviation of each xi from its own sample mean x bar by the deviation of the corresponding yi from its sample mean y bar, and this sum is divided by n minus one.

If you compute the covariance to measure the strength of the linear relationship between the monthly spent and the household income, what does the covariance of S of xy come out to be for Hanumantha's case? It comes to be equal to 16, 109. And if we compute the covariance to measure the strength of the linear relationship between monthly spend and the household size, the covariance S of xi comes out to be 1, 610.

How do we interpret covariance? Yes, let us discuss that. Consider this figure where we have a rectangular region with data points corresponding to x around the horizontal or x axis and data points corresponding to y along the vertical or y axis. Let us draw a vertical dashed line representing x bar and a horizontal dashed line representing y bar.

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Then these two lines divide the rectangle into four quadrants. Points in quadrant 1 correspond to points where xi is greater than x bar and yi is greater than y bar. Points in quadrant 2 correspond to points where xi is less than x bar, but yi is greater than y bar. Points in quadrant 3 correspond to points where x corresponds to points where xi is less than x bar and yi is less than Y

bar. Points in quadrant four corresponds to points where xi is more than x bar and yi is less than Y bar. So the value of xi minus x bar times yi minus Y bar must be positive for the points in quadrants one and quadrant three and the value of xi minus x bar times yi minus y bar must be negative for the points in quadrants 2 and 4.

If the value of covariance s of x y is positive, then the points with the greatest influence on s of x y must be in quadrants 1 and 3, and hence a positive value of s of x y indicates a positive linear association between x and y. That is, as the value of x increases, y increases. Yes, and if the value of S of x, y is negative, then the points with greatest influence on S of x, y are in quadrants 2 and 4.

Hence, a negative value of S of x, y indicates a negative linear relationship between x and y. That is, as the value of x increases, the value of y decreases. Yes. And finally, if the points are evenly distributed across the four quadrants, then the value of S of x, y will be close to zero, indicating that there is not much of a linear association between x and y.

Now, note that if the two variables x and y were the same, that is, if we found the covariance of x with itself, what do we get? Covariance of x with x. Which is? Well, we get the variance of x. Oh, yes, of course. Yes, and in that sense, covariance is a generalization of the notion of variance that we saw earlier.

One problem with the use of covariance as a measure of a strength of linear relationship is that the value of the covariance depends on the units of measurement of x and y. For example, in Hanumantha Pai's case, the covariance between the annual income of the household and the monthly credit card spent.

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Clearly, the strength of this relationship should be the same. Yes, whether we measure the income in rupees or in lakhs. Yes, but however, measuring the income in rupees would give us much larger values of xi minus x bar than if you were to measure these incomes in lakhs. And thus, we would obtain a larger value for the numerator, xi minus x bar, times yi minus y bar and hence a larger value of the covariance.

Oh yes, it is true. When in fact the relationship does not change.

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Correlation

A measure of relationship between two variables that is not affected by the units of measurements of x and y is the correlation coefficient. We will see that measure next. For a sample data, the Pearson product moment correlation coefficient is the most popular correlation measure used. Oftentimes, this correlation coefficient is referred to as the simple sample correlation coefficient.

This correlation coefficient R of x, y is equal to the sample covariance S of x, y divided by the product of the sample standard deviation of x times the sample standard deviation of y. Let us now compute the sample correlation coefficient for Hanumantha's example. We can compute the standard deviation of the annual household income.

What is it? It comes to be equal to 13. 74 lakhs. Yes, and we can calculate the sample standard deviation of the monthly credit card spend, and that comes out to be equal to 1454 rupees. Then, the sample correlation coefficient between the monthly credit card spend and the household income is given by the covariance divided by the product of the standard deviations, which is 16, 109 divided by 13.

74 times 1454. And what does that come out to be? It is equal to 0. 81. Yes, and that indicates a strong linear association between annual income and monthly spend. What about household size and monthly spend? We can also compute the sample standard deviation for household size. What does that come out to be?

It comes to be 1. 76. Yes, and we know that the sample standard deviation of the monthly credit card spend is equal to 1454 rupees. Then, the sample correlation coefficient between the monthly credit card spend and the household size is given by the covariance divided by the product of the standard deviations, which is 1610 divided by 1. 76 times 1454. And what does that come out to be? It is equal to 0. 63. Yes, and that suggests a positive linear association, but not as strong as the association that household income had. Let us now talk a little bit about the interpretation of the correlation coefficient that we just computed.

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If we had sampled data with pairs x1, y1, x2, y2, and so on till xn, yn, that exhibits a perfect positive linear correlation. Then, we would be able to draw a straight line joining the various points, and the value of the sample correlation coefficient would be equal to 1. In general, it can be shown that if all the points in a data set fall on a positively sloped straight line, then the value of the sample correlation coefficient would be plus 1.

So a sample correlation coefficient of plus 1 corresponds to a perfect positive linear relationship between x and y. Yes. And moreover, if the points on a data set fall on a line, a straight line, having a negative slope, then the value of the sample correlation coefficient would be minus 1. So, a sample correlation coefficient of minus 1 corresponds to a perfect negative linear relationship between x and y.

Yes. So, now suppose that the data has a positive relationship between x and y, but the relationship is not perfect. In this case, the value of the correlation coefficient r of x, y will be between 0 and 1, indicating that the points on a scatter diagram are not all on a straight line. Does the reverse happen in case of negative correlation?

Yes, when it is perfectly negatively correlated, r of x y is minus 1. As this association gets weaker, the correlation coefficient takes values between 0 and minus 1. And as the points move away from a perfect linear relationship, positive or negative, the value of r of xy, the correlation coefficient will become smaller and smaller.

And when the value of the correlation coefficient is equal to zero, it suggests that there is no linear relationship between x and y, and the value of r of xy is close to zero. That indicates a weak linear relationship. Now in Hanumantha's data, we can see that the correlation of the monthly spend on the credit card and the household income is 0. 81.

So we can conclude that there is a strong positive linear relationship between monthly spend and household income. Specifically, an increase in the household income is associated with an increase in the credit card spend. However, we should know that correlation only provides a measure of linear association.

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Correlation is not causation. A high correlation between the two variables does not necessarily mean that a change in one variable will cause a change in the other variable. For example, in Hanumantha's case, it is not necessary that just because the household has higher income, they should also spend higher on their credit cards.

Note that because a correlation coefficient only measures the strength of a linear relationship. between two quantitative variables, it is possible for the correlation coefficient to be near zero, suggesting that no linear relationship exists. But the relationship between these two variables could be non linear.

For example, if you look at a scatter diagram that is shown, that shows the relationship between the amount spent on utilities against the temperature in the outside environment, we would expect a non linear relationship as you see. The linear correlation coefficient for this data would be close to zero, which suggests that there is no linear relationship between the variables.

However, the scatter diagram does provide us a strong evidence of a nonlinear relationship. We can see that when the outside temperatures are very high or very low, the amount spent on the utility is high than when the outside temperatures are more moderate. The Pearson correlation coefficient that we saw assesses only the linear relationship between two quantitative variables.

What happens if one or more of the variables are not quantitative? In these cases, there are other correlation coefficients, such as the Spearman rank correlation coefficient and others that can be used to assess the relationship between variables, where at least one of the variables is either nominal or ordinal.

We will discuss them later.

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Covariance and Correlation

Before we go further, let us talk a bit about the computation of covariance and correlation in Excel. Of course, as Ashwini and Tejas did here, you could always take the two variables x and y, compute x bar and y bar for the two sets of variables, and then compute xi minus x bar times yi minus y bar for each value of i ranging from 1 to n, and then compute the covariance using the formula that we just saw.

Similarly, we can compute the correlation by first computing the covariance and then dividing it by the standard deviation of the variable x and the standard deviation of the variable y. However, Excel has inbuilt functions that directly computes the sample covariance and correlation. Oh, that is nice.

Yes. And in Excel, there is a function called covariance. s that computes the sample covariance. It requires two inputs, Array1 and Array2, and if you select Array1 corresponding to x and Array2 corresponding to the y variable, then, the covariance. s function directly gives you the covariance between x and y.

Similarly, Excel also has a function called corral that computes the correlation. It also requires two inputs, array1 and array2. If you select array1 corresponding to x and array2 corresponding to the y variables, you then the function corral directly gives you the correlation between x and y. If you apply these functions to Hanumantha's credit card data set, then, you get the covariance between monthly spend and household income to be 16, 109, and the correlation between the monthly spend and household income to be 0.

806. Similarly, if you use these functions, you will get the covariance between the monthly spend and the household size to be 1, 610. And the correlation between monthly spend and household size to be 0. 629. These are the values that we had got earlier as well. Here, I want to emphasize that the correlation that we are calculating, the Pearson product moment correlation coefficient measures the linear association between two variables.

It does not measure any nonlinear association that might exist between two variables. To emphasize this point, let us consider a special example where there are six pairs of points x and

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y. Let x take values of 3, 2, 1, minus 1, minus 2, and minus 3, respectively. and let the corresponding values of y be defined as 1 over x square, giving us values of 1 over 9, 1 over 4, 1, 1, 1 over 4, and 1 over 9, respectively.

You see these values in the spreadsheet. Now, if you use the Excel functions to compute the covariance and correlation, between x and y, they come out to be both equal to 0, suggesting that x and y have no association. But clearly, we know that there is an association, and the particular function defining this association is y is equal to 1 over x squared.

So notice that in this example, that the correlation and the covariance functions are only measuring the linear association between the variables. So the fact that they are both 0 does not necessarily imply that there is no association at all. In fact, it only implies that there is no linear association between x and y.

We can classify correlation into explainable correlation and spurious correlation. If the correlation between two variables can be explained using logical reasoning, we call it explainable correlation. Otherwise, it is called spurious correlation. One of the major problems with correlation is the possibility of spurious correlation between two variables.

In many cases, spurious correlation is caused due to a third variable called as a hidden or latent variable that influences both of these variables. Let's take a look at a few examples. Crime rate versus ice cream sales. There are published studies in the literature that report that the sales of ice cream and crime rates are positively correlated.

Obviously, ice cream is not driving crime rates. So in this case, there is a hidden variable, which is temperature. Ice cream sales increased during the summer season. The increase in the crime rate in summer is because people are going on vacation and leaving their homes empty as vulnerable targets. Very strange.

Yes. Here is another example. The number of doctors is positively correlated with the number of deaths in villages. Studies have reported that as the number of doctors increase, the number of deaths in villages are also increasing. Why would that be? Well, we can be sure that the doctors are not causing an increase in the deaths.

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What is happening in this case is that there is a need for more doctors in villages, and the public health in villages is poor, causing more deaths in villages. Third example, divorce rate in the state of Maine and the per capita consumption of margarine are correlated. There have been studies that have documented that the divorce rate in the state of Maine in the United States is was reportedly highly correlated with the per capita consumption of margarine.

Based on data collected between 1999 and 2009, the correlation coefficient was computed to be equal to 0. 99. That is high. And what is the hidden variable? Well, this is spurious correlation, as researchers have not been able to find any hidden variable that can explain this high correlation. Here's one more.

Scotland theory. This theory was proposed by the economist George Taylor in 1926. He claimed that the average length of women's skirts and the economy were correlated. According to his theory, the skirts get shorter when the economy is doing well, and they get longer when the economy is doing poorly.

The reason for longer skirts being in fashion during an economic downturn, according to Taylor, was because women could no longer afford stockings. That is really a weird study. Yes, indeed. But clearly, as you can see, there can be several instances of correlation that could be classified as either explainable or spurious.

And so we must be careful in interpreting correlation as shown by the data. The expressions for covariance and correlation that we have seen have been using sample data. Therefore, they are often referred to as sample covariance and sample correlation. Now, like we did for sample mean versus population mean and other population parameters, we could argue that the covariance and correlation we see in the sample data are only point estimates of what we could expect in the population from which this sample was drawn.

Although there are expressions for covariance and correlation in the population, they can only be used if you have the entire data for the population. And in cases like we are interested, where you have only data for a single sample from a population, we would like to know what would be the minimum value of a correlation coefficient obtained from sample data before it can be

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considered to be statistically significant enough to claim that there is correlation in the population too.

That is statistical inference, right? Yes, Tejas, and how did we answer such questions for other parameters? By hypothesis test. Yes, and as you would expect, the answer to this question would also follow from a hypothesis test like we have done for all such inferences about a population parameter using sample data.

But we will not get into the details of these hypothesis tests here, but you should know that the underlying test uses a t distribution and is similar to what we did in the previous module.

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Simple Linear Regression – I

Let's now talk about linear regression. Regression is one of the most important techniques used in predictive analytics. In regression, the outcome or a dependent variable is a continuous numeric variable. Regression is also considered as part of supervised learning algorithms. That is because a regression model requires knowledge of both the dependent and the independent variables from the training data.

A little bit of history before we get in to the details. The statistical methods used in study ing the relationship between two variables was first employed by Sir Francis Galton. Galton was interested in studying the relationship between heights of parents and heights of their children.

To understand the relationship, Galton collected the data of 905 adult children from 205 families. Galton then transmitted the female heights into their male equivalents by multiply ing the female heights by a factor of 1. 08 to account for the difference in male and female heights. Then he observed that when the mid parent height, that is the average height of both the parents, was higher than the mean value for that generation, their children tended to be shorter.

However, when the parents were shorter than the mean value for the generation, their children tended to be taller. So in simple terms, Galton found that shorter parents had taller children and taller parents had shorter children. Interesting finding. Now I know why I'm taller than my parents. And I know why I am shorter than my parents. Wait, the story is not over.

Galton coined the term regression, since the heights of the taller parents were regressing towards the mean value of the heights of the offspring population. Francis Galton probably never realized that the term that he coined would become one of the most important tools in Analytics and Economics. In regression terminology, the variable being predicted is often called as the dependent variable.

The variable or variables being used to predict the value of this dependent variable are called the independent variables. Sometimes the dependent variable is often known as the response variable or the outcome variable. And the independent variables are often called as the predictor variables or the explanatory variables or features.

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Notation wise, it is common to use y to denote the dependent variable and x to denote the independent variable. For example, in analyzing the effect of advertising expenditures on sales, a manager may want to predict sales and consider sales as the dependent variable and the advertising expenditure as the independent variable that is used to predict the sales.

In this module, we will consider the simplest type of regression analysis involving one independent variable and one dependent variable. We will also assume that the relationship between the variables are approximated by a straight line, and this kind of regression is known as simple linear regression.

Regression analysis that involves two or more independent variables is called multiple regression analysis, and we will cover those in the subsequent modules. One of the functional forms of the simple linear regression is given as y is equal to beta 0 plus beta 1 times x plus epsilon. For a data set with n observations, x i commy y i, where i takes value 1, 2, all the way up to n.

The functional form can be written as y i is equal to beta 0 plus beta 1 times x i plus epsilon i. Here, y i is the value of the ith observation of the dependent variable in the sample. x i is the value of the ith observation of the independent variable in the sample. And epsilon i is the random error, also known as the residual or the unexplained variation.

Beta 0 and beta 1 are the regression parameters or the regression coefficients. Simple linear regression attempts to predict the changes or variations in the values of the dependent variable y using the knowledge of the explanatory variable x. Therefore, the equation beta 0 plus beta 1 times x i gives us the predicted value of y for a given value of x i whereas the term epsilon i is the error in the prediction of the values of y i.

In fact, beta 0 plus beta 1 times x i is also known as the conditional expected value of y i for a given value of x i. Now, Ashwini and Tejas, tell me this, if we write y i is equal to beta 0 plus beta 1 times xi square plus epsilon i, is it linear regression or non linear regression? Non linear of course. And if I wrote y i is equal to beta 0 plus beta 1 times the log of xi plus epsilon i.

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Again, non linear regression. No, that's not true. Both of them are examples of linear regression. You know why? The linearity condition in linear regression is defined with respect to the regression coefficients of beta 0 and beta 1 and not with respect to the explanatory variables in the model. For example, the functional form y i is equal to beta 0 plus beta 1 times xi square plus epsilon i is a linear regression model.

So is the functional form y i is equal to beta 0 plus beta 1 times the log of xi plus epsilon i. Because in both of these cases, the relationship of the dependent variable y i with respect to the parameters beta 0 and beta 1 are both linear, although the relationship between y i and xi's are not linear.

On the other hand, regression relationship such as y i is equal to beta 0 plus e raised to beta 1 times xi plus epsilon i, would be an example of a non linear regression model. Since here the relationship between the dependent variable y i and the regression coefficient beta 1 is non linear. In this case, estimating the regression coefficients will involve solving a system of non linear equations, whereas in the case of simple linear regression, estimating the regression coefficients involves solving a set of linear equations.

Many times, non linear regression relationships can be converted into linear regression models by applying suitable transformations. A simple linear regression model is developed to understand how the value of a key performance indicator is associated with changes in the value of independent variables.

We have already seen the example of Hanumanthapa in Canara Bank, where he is interested to know how the monthly spend on credit cards might be a function of either the size of the household or the annual income of the households. As another example of simple linear regression, government might want to understand the impact of specific policies or initiatives

on the wages of the labour force in different sectors of the country. Here's another example. A restaurant might like to know what is the relationship between customer waiting times after placing an order and their popularity ratings. Tejas and Ashwini, can you think of some examples? A hospital may be interested to know how the cost of treatment depends on the weight of the patient.

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Overweight or not? Yes. An e commerce company like Amazon may want to understand how the number of visits to their website relates to sales revenue.

Yes, both are good examples. Here is another one. A manufacturing company may be interested in understanding the relationship between their sourcing decisions and their supply chain costs. Here's one more. Banks might like to understand the impact of an unemployment rate on the percentage of non performing assets.

Of course, these are just a few examples to give you a sense of where simple linear regression might be used. Let us now look at the different steps that are involved in building a simple linear regression model. The first step is to collect and extract data. Now data collection, of course, as we know, can be time consuming and expensive.

Data can be collected through existing sources, and that is known as collecting of secondary data. Or data could be obtained through a primary data collection mechanism, which involves interviews or surveys. The second step is to pre process this data collected. Pre processing the data sometimes involves deriving new features from existing features.

For example, if you recall, Francis Galton multiplied the height of every female by 1. 08 to make it comparable to the height of males. This is an example of preprocessing of data before a model is built. It is essential to investigate the quality of the data during this pre processing step. This might involve removing outliers or dealing with missing data.

Descriptive statistics and data visualization are usually helpful in this pre processing step. The third step in the process is to divide the data into a training and validation data sets. You mean we split or partition the available data? Yes, that is right. The proportion of the training data set is usually around 75 percent and the remaining data is used as validation data.

Subsets of the data may be created using random sampling procedures. The main idea is to use the training data to build multiple models and then test these models on the validation data set. However, there is no guarantee that the best performing model in the training data will also perform best in the validation data.

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What happens in that case? Then, there are several techniques that are used to combine model building and validation in an efficient manner. The fourth step, compute descriptive statistics before model building. Descriptive statistics on the training data help us understand the variability in the data.

They also help us with visualization of the data. You mean box plot or scatter plots? Yes, and they reveal if there are obvious relationships between the variables that should be incorporated into the model. The fifth step is to define the functional form of the relationship. For example, we may have a functional form that y is equal to beta 0 plus beta 1 times x or y is equal to beta 0 plus beta 1 times x squared. We would need to decide whether one of these two forms are better suited to represent the relationship as shown by the data. It might be simple or multiple linear regression. Yes, that is true. And the sixth step is to estimate the regression parameters themselves.

Once the functional form is specified, the next step is to estimate the regression parameters. Usually, the method of ordinary least squares or OLS is used to estimate these regression parameters. What is ordinary least squares method? We will see this in more detail pretty soon, but the main idea behind ordinary least squares is to find regression parameters such that the sum of square distances between the actual observations in the sample and the regression line are minimized.

Ordinary least square method provide the best linear unbiased estimate of the regression parameters. The seventh step is to perform regression model diagnostics. Before a model can be deployed, it is necessary that the regression model created is validated for all assumptions, including the definition of the functional form.

If the model assumptions are violated, then corrective actions must be taken. The eighth step is to validate the model using the validation data set. One of the major concerns here is overfitting. What is overfitting? Overfitting is when the model performs very well in the training set, but it performs very poorly in the validation data set.

Therefore, it is important that the model performance is consistent. The ninth step is to decide on the model deployment. Actual use of the model in decision making. Yes, this involves

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generating actionable items or business rules that can be used while using the model. Of course, the model can be deployed only when its performance is acceptable in the validation dataset.

An ideal model should have low bias and low variance. Yes, that about summarizes the main steps in model building. It sure does give us a lot to think about. Yes, but in subsequent sections and parts of this module, we will examine all of these pieces of model building in more detail. So let's get into the details.

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Simple Linear Regression – II

Let us start by understanding the regression equation in some more detail. Recall that the functional form can be written as yi is equal to beta 0 plus beta 1 times xi plus epsilon i, where yi is the value of the ith observation of the dependent variable in the sample. And xi is the value of the ith observation of the independent variable in the sample.

And epsilon i is the random error also known as a residual or the unexplained variation. Beta 0 and beta 1 are the regression parameters or the regression coefficients. To understand this regression equation in more detail, let us look at Hanumanthapa's example again. In this case, the population consists of all his customers.

For every customer of the bank, there is a value of x, the annual income, and a corresponding value of y, the monthly spend on the credit card. The equation then describes how the monthly spend y is related to the annual income x along with an error term. However, the population in Hanumantha's case can also be viewed as a collection of sub populations.

One for each distinct value of x, or annual income. For example, one subpopulation might consist of all the customers that have an annual income of about 20 lakhs. The second subpopulation might consist of all customers that have an annual income of about 30 lakhs. And a third subpopulation might consist of all customers that have an annual income of 40 lakhs, and so on.

Thus, a distribution of the Y values is associated with customers having an annual income of 20 lakhs. A distribution of the Y values is associated with customers having an annual income of 30 lakhs and a distribution of the y values is associated with customers having an annual income of 40 lakhs and so on.

Now each distribution of these y values will have its own mean or expected value. Then the equation y i equal to beta 0 plus beta 1 times x i plus epsilon i then describes how the expected value of y or E of y is related to x. Therefore, the interpretation of our regression equation is that the expected value of y, E of y is equal to beta 0 plus beta 1 times x.

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The error term is gone when we take the expectation. Yes, that is right. Now, if the values of the population parameters beta 0 and beta 1 were known, then we could use the equation of the expectation of y equal to beta 0 plus beta 1 times x to compute the mean value of y for any given value of x.

However, in practice, the parameter values beta 0 and beta 1 are not known. They must be estimated using sample data. That is the training data set we talked about. Yes, and from sample data, sample statistics denoted by B0 and B1 instead of the beta 0 and beta 1 are computed as estimates of the population parameters beta 0 and beta 1.

Substituting the values of these sample statistics B0 and B1 instead of the beta 0 and beta 1 in the regression equation, then gives us an estimated regression equation. This estimated regression equation for simple linear regression is then written as y hat is equal to b zero plus b one times x. We can see a summary of this estimation process for simple linear regression summarized in this figure.

The graph of the estimated simple linear regression equation is known as the estimated regression line. For this line, B0 is the y intercept of this line, and B1 is the slope of this line. We will see how to use ordinary least squares method to compute the values of B0 and B1 in the estimated regression equation.

We will see how to use ordinary least squares method to compute the values of b0 and b1 in the estimated regression equation. Also, y hat is the point estimator of the expected value of y, or the mean value of y, for a given value of x. What this implies is the following. To estimate the mean value of monthly spend for all customers with an annual income of about 20 lakhs, Hanumanthappa would substitute the value of 20 lakhs for x

in the estimated regression equation. However, in some cases, Hanumantha might be more interested in predicting the monthly spend for customers with a certain income. For example, suppose Hanumantha is interested to predict the monthly spend for customers with an annual income of 25 lakhs. Then, the best predictor for y for the given value of x equal to 25 is also given by y hat, which is equal to b zero plus b one times x.

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So, to predict a monthly spend of customers with an annual income of 25 lakhs, Hanumantha would substitute the value of 25 lakhs in the estimated regression equation. Okay, now let us see how we compute these values of b0 and b1 using the ordinary least squares method. The least square method is a procedure for using sample data to find the estimated regression equation.

We already saw through a scatter diagram of Hanumantha's data that the annual income on the x axis and the monthly spend on the y axis, the relationship appears to be linear and increasing. In other words, a monthly spend appears to be higher for higher values of annual income. The least square method uses sample data to find the values of b0 and b1 such that it minimizes the sum of squared deviations

between the observed values of the dependent variable y i and the predicted values of this dependent variable y i hat. The criteria for the least square method is given as follows. Find the values of b 0 and b 1 such that it minimizes the sum of the squares of y i minus y i hat for all points 1 through n in the training data set.

Here, y i is the observed value of the independent variable for the i th observation and y i hat is the predicted value of the dependent variable for the i th observation. Tejas and Ashwini, do you remember differential calculus? Oh no, are we going back to Calculus? Well, yes and no, but don't worry. Differential calculus can be used to show and find the values of b0 and b1 that minimize the expression of the sum of least squares.

Using calculus, we can show that b1 is given by the following expression. b1 is given by the summation of the product of xi minus x bar times yi minus y bar over all the values of i in the numerator, divided by the summation of x i minus x bar square in the denominator. The value of b 0 is then given by y bar minus b 1 times x bar.

In this expression, x i is a value of the independent variable for the ith observation, y i is a value of the dependent variable for the ith observation. x bar and y bar are the mean values of the independent and dependent variables respectively and n is the total number of observations. If we did these calculations for Hanumantha's data set, we would get x bar to be equal to the summation of all the xi's over n, which comes out to be equal to 32. 89 lakhs. We would get y

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bar to be equal to the summation of yi's over all n. We would get y bar to be equal to the summation of all the yi's over n, which would come out to be equal to 3, 852. Then we would get the value of b1 to be equal to 85. 27. And the value of b0 would come out to be equal to 1047. 37. Therefore, the estimated regression equation would be that y hat is equal to 1047. 37 plus 85. 27 times x.

The slope of this estimated regression equation is b1 and is equal to 85. 27 and that is positive. This implies that as the annual income increases, the monthly spend also increases. In fact, we can conclude that an increase in an annual income by one lakh results in an increase in the monthly spent by 85. 27. However, one should be cautious about making predictions based on regression values of the independent variable outside the range of values in the data that are used to estimate the regression equation in the first place. We cannot be sure that this relationship that we have just estimated is valid outside the training data.

If you believe that the least squares estimated regression equation adequately describes the relationship between x and y, then it would seem reasonable to use this estimated regression equation to also predict the value of y for a given value of x. For example, if you wanted to predict the monthly spend for a customer who has an annual income of 35 lakhs, then, we would compute the y hat to be equal to 1047. 37 plus 85. 27 times 35, and that would come out to be equal to 4032. Therefore, we would predict a monthly spend of 4032 for this customer. Next, we will discuss methods for assessing the appropriateness of using this estimated regression equation for estimation and prediction.

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SLR – OLS

We saw that we can write y hat is equal to b zero plus b one times x. And then using calculus, we can find b zero and b one to minimize the sum of least square errors. We said that b one is given by the following expression. The summation of the product of x i minus x bar times y i minus y bar over all values of y in the numerator divided by the summation of x i minus x bar square in the denominator. The value of b 0 is then given by y bar minus b 1 times x bar. We can see the computation of b 1 and b 0 in a spreadsheet right here. In column G, we have the values of annual income x. In column H, we have the values of the monthly spent, y.

Then in column I, we compute the values of xi minus x bar for each of the 55 data points that we have. And in column J, we compute the values of y i minus y bar for each of the 55 data points. Then in column K, we compute the value of xi minus x bar times yi minus y bar. And finally, in column L, We have the values of x i minus x bar square. Then we know that the value of b 1 is given by the summation of the values in column K divided by the summation of the values in column L.

And that comes out to be 85. 27. We also have that y bar minus b 1 times x bar is equal to b 0. We take the average value of the values in column H minus 85. 27 times the average of the values in column G and we get 1047. 37. Therefore our equation y is equal to b0 plus b1 times x or in other words y hat is equal to 85. 27 plus 1047. 37 times x. Do we have to do this manually for every data set? Not really. There are a few options in Microsoft Excel itself. We will see some of them now. For example, Microsoft Excel has a convenient option of determining the values of B0 and B1 by plotting a trend line on a scatter plot.

If you take the scatter plot of the monthly spend on the y axis and annual income on the x axis, and click on any of the data points on the plot, and then right click, you will get a menu option which says add a trendline. With that option, if you choose the linear trendline option, and then check on the boxes corresponding to display equation on the chart and display r square value on the chart, you will get a linear trendline as well as the equation with b0 and b1 values that we just found. Oh, Excel you are my savior once again.

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Actually Ashwini, we'll see that in a bit excel can do a whole lot more for linear regression. Anyway, along with the scatter plot and the trend line and the values of B zero and B one, we can also compute a value of R Square. Now, we haven't discussed the concept of R Square yet. And that's what we're going to discuss next.

Before we use a regression model for decision making, it is important to ensure its validity and goodness of fit. How do we ensure its goodness of fit? There are several measures that we can use to validate simple linear regression models. One of the first things we do is to compute the coefficient of determination or R squared.

For Hanumantha's data set from the Canara Bank example, we found that the estimated regression equation is y hat is equal to 1047. 37 plus 85. 27 times x to approximate the linear relationship between monthly spend y and the annual income x. The question is, how well does the estimated regression equation fit the given data?

Is there a goodness of fit measure? Yes, there is. And we will show that the coefficient of determination provides a good measure of the goodness of fit for the estimated regression equation. The coefficient of determination, r square. measures the percentage of variation in y that is explained by the model beta 0 plus beta 1 times x.

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SLR - Coefficient of Determination

The simple linear regression model can be broken down into explained variation and unexplained variation. Recall that y is equal to beta zero plus beta one times x plus epsilon i. The beta zero plus beta one times x i part of the model tries to explain part of the variation in y. The epsilon i term is the variation in y that is not explained by the model.

I did not think of this equation in that way. Yes, so let us look at this a little bit more carefully. For the i th observation, the difference between the observed value of the dependent variable yi and the predicted value of the dependent variable yi hat is called as the i th residual. This i th residual represents the error in using yi hat to estimate yi.

Thus, for the i th observation, the residual is given by yi hat minus yi. For example, in Hanumantha's data set, corresponding to an annual income of 22 lakhs, the actual monthly spend on the credit card was 2429. However, the predicted value of the monthly spend y hat is 2923. So, the residual is y i minus y i hat, which is 2429 minus 2923, which comes out to be minus 494.

Similarly, we can compute the residual for the other data points. I thought the least squares method was minimizing these residuals. Yes, Tejas, and the sum of squares of these residuals or errors is the quantity that is minimized by the least square method. This quantity is also referred to as the sum of squares due to error and is denoted by SSE. The formal definition of SSE is the summation of yi minus yi hat square for all values of i. The value of SSE is a measure of the error in using the estimated regression equation to predict the values of the dependent variable in the sample.

In the spreadsheet, you can see SSE to be computed in column Y to be equal to 40017408. That is a big number, but it is relative, right? Now how do we know? We will see, soon. Now suppose you were asked to develop an estimate of the monthly spend without the knowledge of the size of the annual income or any related variables. What would you do? Without giving us any values of x, only values of y. If I have to predict only using values of y, I would just compute the average as my prediction.

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Yes, exactly. We would use the sample mean of the monthly spend as an estimate or a prediction of the monthly spend. In Hanumantha's dataset, the average value of YI is Y Bar and is equal to 4 0 5 1. Now we compute the sum of square deviations obtained by using this prediction, the sample mean Y bar to predict the value of the monthly spend for each observation, YI. The difference between the actual value in the sample, YI minus Y bar provides a measure of the error involved by using Y Bar to predict the monthly spend. Is there a term for this quantity? Yes. The corresponding sum of squares is called the total sum of squares or SST.

The formal definition of the total sum of squares SST is given by, the summation of y i minus y bar squared over all values of i. In Hanumantha's data set, SST comes out to be equal to 114197923. Wow, that is an even bigger number than SSE, that we just computed. Yes, it helps to think of SST as a measure of how well the observations cluster about the value of y bar equal to 4 0 5 1 and SSE as a measure of how well the observations cluster about the line y hat is equal to b 0 plus b 1 times x.

Now one more quantity to compute before we can say something about the model. What are we computing next? To measure how much y hat values on the estimated regression line deviate from y bar, another sum of squares is computed. This sum of squares is called as the sum of squares due to regression, and this is known as SSR.

The formal definition of the sum of squares due to regression is the summation of y i hat minus y bar square over all values of i. In Hanumantha's data set, we can compute the values of SSR to be equal to 74180516. I guess the SST, SSE and SSR will tell us about the quality of the model. Yes, exactly. So Tejas and Ashwini, we should expect from this discussion that the three sum of squares SST, SSE and SSR that we have defined and computed are related.

And indeed, the relationship amongst these three sums of squares is one of the most important results in statistics. It can be proven mathematically that the sum of squares of the total variation, or SST, is equal to the sum of squares of the explained variation, SSR, plus the sum of squares of the unexplained variation, SSE.

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That is the summation of y i minus y bar square is equal to the summation of y hat minus y bar square plus y i minus y bar square, where the summations are over all values of i. Now, can we verify this for Hanumantha's case? Yes, let's add the two big numbers to see if we get the third big number. Yes, and in Hanumantha's data set, we can see that SST is equal to 114197923.

And that is equal to SSR, which is 74180516 plus the SSE which is 40017408. Yes it does. Interesting. Now let us see how these three sums of squares SST, SSR and SSE can be used to determine a measure of goodness of fit of the estimated regression equation. Note that the estimated regression equation would provide a perfect fit if every value of the dependent variable y i happens to be exactly on the estimated regression line.

And in this case, y i minus y i hat would be equal to zero for each and every observation. And that would give us an SSE equal to zero. But now, since SST is equal to SSR plus SSE, then for a perfect fit, SSR must be equal to SST. And what will be the ratio of SSR over SST? I think it will be one. Yes. And as the fit of the estimated regression equation gets worse, we will see larger values of SSE other than this minimum value of zero.

Now, what can be the largest value of SSE? The largest value of SSE would be corresponding to the poorest fit, and this would occur when SSR is equal to zero. Yes, and in this case, SSE would be equal to SST. And therefore, the ratio of SSR over SST can take values between 0 and 1. This ratio takes a value of 1 when SSE is 0 and this ratio takes a value of 0 when SSE is equal to SST.

So this ratio then of SSR over SST is used to evaluate the goodness of fit of the estimated regression equation. Is there a name for this ratio? Yes, and this ratio is known as the coefficient of determination and is denoted by R squared. The formal definition of the coefficient of determination is R squared is equal to SSR divided by SST.

For Hanumantha's data set, we can see that SSR divided by SST is equal to 0. 650. Yes, and if we express this coefficient of determination as a percentage, then, r square can be interpreted as the percentage of the total sum of squares that can be explained using the estimated regression equation.

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Therefore, in Hanumantha's example, we can conclude that 65 percent of the total sum of squares can be explained by using the estimated regression equation of y hat is equal to 1047. 37 plus 85. 27 times x. In other words, 65 percent of the variability in the monthly spent can be explained by the linear relationship between annual income and the monthly spent.

So, the higher the r square value, the better is the fit of the regression model to the data. Yes, exactly.

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SLR - Correlation Coefficient vs Coefficient of Determination

Next, we look at a related concept known as the correlation coefficient. Tejas and Ashwini, if you recall when we started this module, we talked about correlation coefficient as a descriptive measure of the strength of the linear relationship between two variables x and y. Yes, the values of correlation coefficient always lie between minus one and plus one.

Yes, and a value of plus one indicates that the two variables are perfectly related in a positive linear sense. That is, all data points lie on the straight line with a positive slope. Yes, and a value of minus 1 indicates that x and y are perfectly related but in a negative linear sense. With all data points lying on a straight line that has a negative slope.

Yes, and the values of correlation coefficient that are close to zero indicate that x and y are not linearly related. If the regression analysis has already been performed, and we already have the coefficients of b zero and b one, and the coefficient of determination r square, then the r square value is related to the correlation coefficient as follows.

The correlation coefficient between x and y is equal to the positive square root of the coefficient of determination multiplied by the sign of b1 obtained from the regression equation. In other words, if the coefficient b1 is positive, then the correlation between x and y is directly the positive square root of the coefficient of determination.

On the other hand, if the coefficient b1 is negative, then the correlation between X and Y is the positive square root of the coefficient of determination multiplied by minus one. In Hanumatha's dataset. For example, the coefficient of determination corresponding to the regression equation Y hat is equal to 1 0 4, 7 0.37 plus 85.27 times X is given by 0.65.

We see that B one is positive. Therefore, the correlation coefficient between the monthly spend and the annual income x will be the square root of 0. 65. Can you tell me what it is? It comes out to be 0. 806. Yes, and with such a sample correlation coefficient of 0. 806, we should expect a strong linear association between the monthly spend and the household income.

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One final point here about the connection between the coefficient of determination and the correlation coefficient. We saw that the sample correlation coefficient is restricted only to linear relationships between two variables. However, the coefficient of determination can be used for nonlinear relationships and for relationships between multiple variables.

Thus, the coefficient of determination that we just saw provides a wider range of applications than we will see in terms of applications when we look at only correlation coefficient. We will see the broader range of these applications. when we look at multiple linear regressions in the next module.

Let's recap what we have done so far. For the case of a simple linear regression, the assumed regression model is y is equal to beta 0 plus beta 1 times x plus epsilon. Then the least square method was used to develop b 0 and b 1, which were estimates of the model parameters beta 0 and beta 1, respectively.

Using those estimates, we developed an estimated regression equation to be y hat is equal to b0 plus b1 times x. Then, using the value of coefficient of determination, r squared, we can measure the goodness of fit of the estimated regression equation. However, even if the value of r square is high, the estimated regression equation should not be used until the appropriateness of the assumed model has been evaluated.

This involves testing the significance of the linear relationships that we are assuming in the model.

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SLR - Assumption of Error Term

When we test the significance of the linear relationship, we make a few assumptions about the error term epsilon. Let us see those assumptions. The first assumption is that the error term epsilon is a random variable with mean or expected value of zero. This implies that since beta 0 and beta 1 are constants,

then the expected value of Y will be equal to beta zero, plus beta one times X. The second assumption is that the variance of Epsilon, the error tone denoted by Sigma Square is the same for all values of X. This means that the variance of Y about the regression line equals Sigma Square and is the same for all values of x.

The third assumption is that the values of the error term epsilon are independent. Since the value of epsilon, the error term for a particular value of x is not related to the value of epsilon for any other value of x, then the value of y for a particular value of x is not related to the value of y for any other value of x as well.

The fourth assumption is that the error term epsilon is normally distributed random variable for all values of x. This assumption along with our earlier assumptions imply that we are assuming that the error term is normally distributed with a mean 0 and variance equal to sigma squared. Now, because y is a linear function of epsilon, y is also normally distributed random variable for all values of x.

This figure that you see here is a good way to see the collective implication of all the model assumptions. Note that the expected value of y will change according to a specific value of x and this expected value will be given by beta 0 plus beta 1 times x. However, regardless of the x value, the probability distribution of epsilon, and hence the probability distributions of y, are normally distributed, each with the same variance.

The specific value of the error at any point depends on whether the actual y is greater or less than the expected value of y. Are these assumptions restrictive? Not really, and in many cases, these are perfectly reasonable, as we will see. In a simple linear regression equation, we have the E of y is equal to beta 0 plus beta 1 times x.

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Now, if the value of beta one is zero, then the expectation of Y is going to be equal to beta zero. In this case, the mean value of Y will not depend on X and we can conclude that X and Y are not linearly related. However, if beta one is not zero, then we can conclude that X and Y do indeed have a linear relationship.

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SLR - Test for Significance

Now, using the training data, we get an estimate of B1. However, the question is whether there is enough evidence in our training data beyond reasonable doubt through B1 or otherwise, that there is indeed a linear relationship between X and Y in the population. Are we going to do a hypothesis test? Yes, but how did you guess?

You started talking about sample to population and phrases like enough evidence, beyond reasonable doubt. Yes, you are paying attention, Tejas. I am impressed. So, we want to test whether there is a significant evidence to claim that beta 1 is not equal to 0. So that becomes an alternate hypothesis and the null hypothesis if beta 1 equals to 0.

Exactly and so if you can reject the null hypothesis then beta 1 is not equal to 0 and a linear relationship exists. Now the natural questions are what is the test statistic in this case, what is its distribution and how do we compute the p value. There are two test statistics that we can use. One uses a t statistic and the other uses an f statistic.

In the case of simple linear regression, they are both equivalent. But when we do multiple linear regression in the next module, you will see the different implication. Both the tests require estimates of sigma square, the variance of the epsilon in the regression model. We will derive the estimate from sample data in two ways, in one way for the t test and in another way for the f test.

Let us look at the t test first. For the t test, we use the fact that that the mean square error, MSE, provides an estimate of sigma squared. What is the mean square error? How is it related to SSE? Every sum of squares has associated with it a number called its degrees of freedom. Now, SSE is computed as the sum of yi minus yi hat squared.

And yi hat is given by b0 plus b1 times x. Therefore, SSE has n minus two degrees of freedom because two parameters beta zero and beta one must be estimated. Thus, the mean square error, MSE is computed by dividing the SSE by N minus two. And MSE provides an estimate of Sigma Square. And sometimes, s square is also used as the notation.

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The standard error of this estimate is the square root of s square, that is, s. Now what do we get for Hanumantha's case? The SSE was 4 0 0, 1 7, 4 0 8. So the mean square error, MSE, will be SSE divided by n minus 2, which is 4 0 0, 1 7, 4 0 8 divided by 53. And what does that come out to be? It comes to be seven lakh fifty five thousand zero four five.

Yes. And s is equal to the square root of seven five five zero four five, which comes out to be eight sixty eight point nine. It can be shown that the sampling distribution of B one is normal with mean equal to beta one, and a standard error equal to sigma divided by the square root of the sum of XI minus X bar square.

We use the S that we just computed instead of the sigma to compute the standard error. We use the S that we just computed instead of sigma to compute the standard error. So, the standard error is 868. 9 divided by the square root of 10201, and that comes out to be 8. 603. The test statistic is b1 minus beta 1 divided by its standard error.

And since we are using s instead of sigma in the standard error, the test statistic has a t distribution with n minus 2 degrees of freedom. It comes to be 7 5 5 0 4 5. Yes, and then to find the p value, we evaluate this test statistic when the null hypothesis is true, which is assuming that beta 1 is equal to 0.

And that means that the test statistic in this case is simply b1 divided by the standard error. We compute this value of test statistic that has a t distribution and compute the p value as we would do for a two tailed test. For Hanumantha's data set, this comes out to be 85. 27 divided by 8. 603, which gives us a t value of 9. 91. And that gives us a p value that is very close to zero. In fact, 1. 1 times 10 raised to minus 13. Just like we had for hypothesis test for mean. Yes, and we can confidently say that there is evidence from the sample data that beta 1 is definitely not equal to 0 and a linear regression model is assumed to be valid.

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SLR - Summary Statistics

The first input you need to select is the y variable, that is the dependent variable. In this case, it would be the monthly spent on the credit card. We need to select the array that consists of the 55 data points corresponding to the monthly spent. The next input is for the x variables. In this case, we have only one independent x variable corresponding to the annual income.

So we select the 55 data points corresponding to the annual income corresponding to the y values that we have just selected. Next, we select a cell range where the output of the regression analysis needs to be pasted. We also select the other options for residuals, residual plots, standardized residuals, and line fit plots.

And if you click OK, we will get a regression output that looks like this. Oh my, that is so cool. That is a lot of numbers. Yes, but we have seen most of these numbers. A lot of these were what we were just calculating. So let us see this output in more detail. In the summary section of the output, we find the R square value and the standard error.

If you recall, for Hanumantha's data set, we had computed the R square value to be equal to 0. 65. We had also commented that the correlation between the monthly spend and the annual income would be the square root of this r square value, which comes out to be 0. 806. Next, we see that the standard error 868. 9 that we just computed, this is the value of s that we estimated using the sample data. The square of this value s square would be the estimate of the sigma square corresponding to the variance of the epsilon used in the regression equation. What is this ANOVA section? Yes, next we see the ANOVA table or an analysis of variance table.

This table summarizes the F test that we just saw to determine whether there is a significant linear relationship between y and x. Now, although we haven't covered the concept of ANOVA or analysis of variance in detail, we have actually talked about all the pieces of information that are summarized in the table.

In the context of the F test that we just discussed. The table shows two sums of squares being computed. The sum of squares regression and the sum of square's residual. The sum of square's

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residual is what we have been referring to as sum of square's error or SSE. You will notice that the sum of square's residual is equal to the value of SSE 4 0 0 1 7 4 0 8 that we computed.

You will notice that the sum of squares regression SSR is exactly equal to the 7 4 1 8 0 516 that we also computed. Dividing by their respective degrees of freedom, 1 and 53 respectively, we get the mean square regression to be equal to 7 4 1 8 0 5 1 6 and the mean square error to be equal to 7 5 5 0 4 5.

The F statistic then comes out to be the ratio of MSR over MSE, which comes out to be 98. 25. And is the significance value of f the p value? Yes, it is. And as you recall, it was fairly close to zero, which allows us to reject the null hypothesis that beta one is equal to zero and conclude that there is enough information in the sample to conclude that the linear relationship between Y and X is statistically significant. Next, in the bottom part of the table, we find the coefficients estimated by the regression analysis, the values of B zero and B one. The intercept value is 1 0 4 7 0.37 and that is the value of B zero and the coefficient. 85.27 is the value of B one.

This implies that the estimated regression equation is given by 1047. 37 plus 85. 27 times x. Now, let's look at the row corresponding to b1. You will notice that a standard error corresponding to annual income to be equal to 8. 6. If you recall, when we did a t test to determine whether or not the coefficient beta 1 is equal to 0 or not.

Yes, we had computed the test statistic to be equal to b1 divided by its standard error. Yes, and the estimated standard error for b1 was computed as s divided by the square root of the sum of xi minus x bar square. And we had computed this value to be equal to 868. 9 divided by the square root of 10201, which came out to be 8. 603. That is, what we see here is the output. Yes, and the test statistic, which was given by B1 divided by the standard error, comes out to be equal to 85. 27 divided by 8. 6. which comes out to be 9. 91.

This is exactly the value that we had calculated. Yes, and then the p value corresponding to this test statistic 9. 91, coming from a t distribution with degree of freedom equal to 53, we saw was fairly close to zero. That allows us to reject the null hypothesis. Yes, and so you can see that all the analysis that we did stepwise can be done directly through the use of this feature in Excel. This is very helpful. If all this can be obtained directly, why did we do this individually?

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Yes, that is a good question, Tejas. And I did this individually so that you know where the outputs provided by this tool pack are coming from. For instance, the same tool pack has a plugin for descriptive statistics that we used in module 1. That plugin gives us the mean, median, and several other descriptive statistics.

However, we still discussed how these were computed, so we understand the output provided by this Excel plugin. The idea here is fairly similar. That being said, when we solve a few more examples in a little bit, we will not do the calculations individually. Instead, we will directly obtain this output and spend more time on interpreting the results.

Okay, now you can see that this inbuilt regression plugin gives us a lot of information about the underlying regression analysis. In fact, there is information below this table and I'm going to discuss them as well.

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SLR - Residual Analysis

Let us now talk about residual analysis. And how, looking at the residual, we can validate the model assumptions. As we said earlier, the residual for the ith value is the difference between the observed value of the dependent variable yi and the predicted value of the dependent variable yi hat. That is, the residual is the error resulting from using the estimated regression equation

to predict the value of the dependent variable. In the Excel output, we see the residuals for Hanumantha's dataset. For each of these 55 observations, we can see the value predicted for monthly spent given by the estimated regression equation. And the value of the residual obtained by comparing the predicted value with the actual value in the data set.

If you recall, we had made some assumptions regarding the error term epsilon in the regression model, y is equal to beta 0 plus beta 1 times x plus epsilon. Do you remember them? We had said that the expected value of epsilon is zero. Yes, and that the variance of the residuals denoted by sigma square should be the same for all values of x.

This property is known as homoscedasticity. We also assume that the values of residuals are independent. Yes, and that the error term epsilon has a normal distribution. These assumptions were vital in being able to justify and conduct the t test and the f test that we just did to check whether the relationship between x and y are statistically significant.

If the assumptions about the error term epsilon do not hold, then the hypothesis test and the significance of the regression relationship may not be valid. The residuals, therefore, provide the best information about the error term epsilon. And hence, an analysis of the residuals helps us to determine whether the assumptions related to epsilon were appropriate.

The good news is that much of this residual analysis is based on examining some graphical plots. Here, we will look at three plots. The graphical plot of the residuals against the independent value of x, a plot of the residuals against the predicted values of the dependent variable y, and a standardized residual plot.

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A residual plot against the independent variable x is a graph in which the values of the independent variable x are represented on the horizontal axis and the corresponding residual values are represented on the y axis. A point is plotted for each residual in this manner. We can see that the analysis toolpack output has provided us a plot of the residuals against the values of the independent variable x.

Before interpreting the results of this residual plot for Hanumantha's data set, it is useful to consider some general patterns that might be observed in any residual plot. Are we looking for any particular types of plots or patterns? Yes, we are usually interested in three patterns that are summarized in this figure.

If you recall, we made an assumption that the variance of epsilon is the same for all values of x and that the assumed regression model is an adequate representation of the relationship between the variables. If that is the case, then the residual plot should give an overall impression of a horizontal band of points such as the one in panel A.

However, if the variance of epsilon is not the same for all values of x, and in particular, if the variability about the regression line is greater for larger values of x, then a funnel pattern such as the one shown in panel B would be observed. In this case, the assumption of constant variance or homoscedasticity is violated.

Another possible residual plot is the one shown in panel C, which has roughly a U shape. Here, we conclude that the assumed regression model is not an adequate representation of the relationship between the variables. A curvilinear regression model or a multiple regression model could be a better choice.

What if a residual pattern does not fall into one of these three categories? Yes, and that can very well happen. Rarely does a residual plot conform exactly to one of these three patterns that we have just discussed. Yet, understanding the difference between these three patterns helps us to critically evaluate a residual plot that we might be looking at for regression model that we have just built.

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So now, what can we say about the residual plot for Hanumantha's dataset? It looks more like a band rather than a funnel or a U shape. And what do you think, Ashwini? Yes, I agree. So our assumptions regarding epsilon seems to hold. Yes, fair enough. Now, the residual plot is one technique to assess the validity of the assumptions of a regression model.

We can also plot residuals against the predicted values y hat instead of the x values. Another form of the residual plot is one where the predicted value of the dependent variable y hat is plotted on the horizontal axis and the residual values are plotted on the vertical axis. As before, a point is plotted for each residual.

Note that the analysis tool pack does not automatically give us this residual plot. But this residual plot can be easily obtained using the outputs provided by the analysis tool pack. Note that the pattern for this residual plot is the same as the pattern for the residual plot for the independent variable x for Hanumantha's dataset.

And we can say that again like we did for the previous residual plot, it has a horizontal, band pattern. So it is not a pattern that would lead us to question the model assumptions. Note that for simple linear regression, both the residual plot against x and the residual plot against y hat provide the same pattern.

However, for multiple linear regression that we will see in the next module, this residual plot with y hat on the horizontal axis is more widely used because of the presence of more than one independent variable. The analysis tool pack also provides us with a standardized versions of the residuals. And as we have seen earlier, a random variable can be standardized by subtracting its mean and dividing its result by the standard deviation.

Now, since we have obtained these residuals using the least squares method, the mean of these standardized residuals is zero. Therefore, by simply dividing the residual by its standard deviation, gives us the standardized residuals. We can then compute the standardized residuals and plot them against the independent variable x.

A plot of the standardized residuals provides insights about the assumption that the error term epsilon has a normal distribution. How does a plot of a standardized residual help us with this?

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If the normal assumption is satisfied, then the distribution of the standardized residuals must come from a standard normal probability distribution.

Thus, when looking at the standardized residual plot, we should expect to see that roughly 95 percent of the standardized residuals should lie between minus 2 and plus 2. Oh yes, the property of the standard normal. Yes, exactly. And if you look at the standardized residual plot for Hanumantha's data set, we see that this is indeed the case.

It is nice that these computations are done by Excel for us. It is quite cumbersome otherwise to do this manually for a dataset. Yes, Tejas, that is correct. Manual computation can be hard, especially if the dataset is large. However, most statistical software, even software like Excel, have inbuilt capabilities to compute these outputs, so that our focus is then on the interpretation of the outputs from regression. Now that we have seen how residual analysis can be used to determine whether violations of the assumptions about the regression model are happening.

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SLR - Outliers and Influential Observations

We will now discuss how residual analysis can be used to identify observations that can be classified as outliers or as being specially influential in determining the estimated regression equation. Some steps need to be taken when such observations are seen. And of course, an outlier is a data point that does not fit the trend shown in the remainder of the data.

If you see an outlier in such a scatter plot, then it requires careful examination. What action do we take? Remove it? Sometimes we do, but outliers may represent erroneous data and if that is the case, they should be corrected. Alternatively, they may represent the violation of the model assumptions and if that is the case, another model should be considered.

Finally, there may just be unusual values that are occurring by chance, and they could very well be retained. Standardized residuals can be used to identify outliers. If an observation deviates greatly from the pattern of the rest of the data, the corresponding standardized value will also have a large absolute value.

Tejas, do you recall that for normally distributed errors, the standardized residuals should be in the range of minus 2 to plus 2 95 percent of the time? Yes, we saw that earlier. Yes, and therefore, data points corresponding to cases where the standardized residuals lie outside the values of minus 2 or plus 2 could be considered as outliers in the dataset.

The question now becomes how do you handle these outliers or whether you decide that it is indeed a valid observation. First, we need to check whether the values are actually valid observations or erroneous data entered in the dataset. If these are erroneous data, then we can correct the value and rerun the regression equation.

Usually, the R square value improves and a new regression equation is obtained when you make such a correction. Now let us talk a little bit about detecting influential observations. Sometimes one or more observations exert a strong influence on the results obtained from simple linear regression. If such an influential observation is dropped from the data set, the slope of the estimated regression line could change and the parameters of the regression model could change as well.

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How do you detect influential observations? Yes, influential observations can be detected using a scatter diagram when only one independent variable is present. For example, an influential variable may be an outlier observation with a y value that deviates substantially from the trend. Alternatively, it may correspond to an extreme value of the independent variable x that is far away from its mean, or it may be caused by a combination of these two phenomenon, a somewhat off trend y value and a somewhat extreme x value. Now, since such influential observations may have a dramatic effect on the estimated regression equation, they must be examined carefully. What do we do with them?

First, we should check to ensure that no error was made in collecting or recording the data. If an error was made in recording the data, it can be corrected and a new estimated regression equation should be developed. Yes, that seems the obvious thing to do. Yes, and if the observation is indeed valid, then such observations could contribute to a better understanding of an approximate model and that can lead to a better estimated regression equation.

In the presence of an influential observation, especially when it is valid. It is recommended to try and obtain data on intermediate values of x to better understand the relationship between x and y.

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SLR - Test of Significance and Residual Analysis

We have gone through the basic concepts related to simple linear regression. We covered the theoretical foundations of simple linear regression and at each step we applied these concepts to Hanumantha's data set. Let's review what we did before we apply these concepts to a few other examples. In simple linear regression, we are trying to establish a relationship between a dependent variable y with a single independent variable x.

The linear regression model is y is equal to beta 0 plus beta 1 times x plus epsilon. The idea is to use the data as a training set and derive estimators of parameters beta 0 and beta 1 which we called as b 0 and b 1. Doing so gives us the estimated simple linear regression equation y hat is equal to b 0 plus b 1 times x.

We use the ordinary least square method to identify b 0 and b 1. The ordinary least square method then tries to minimize the sum of yi minus yi hat square, which we also called as the sum of squares error, or SSE. Then, to determine how well the estimated regression equation explains the data, we computed the coefficient of determination, r square.

To compute r square, we computed the total sum of squares, SST which was the sum of yi minus y bar square. And what did we learn about SST? We saw that SST is equal to the sum of SSE and SSR. Yes, where SSE is given by the summation of yi minus yi hat square and SSR is given by the summation of yi hat minus y bar square.

Using the sum of squares, SST SSR and SSE, the coefficient of determination was then defined. Do you remember what it was? R square is equal to SSR by SST. Yes. And we saw that this can take a value between 0 and 1. When higher the value, the better is the ability of the regression model to explain the variation in the observed data.

We also saw that the correlation between y and x can be obtained by the square root of this coefficient of determination multiplied by the sign of the coefficient of b1. Then what did we do? A couple of hypothesis tests, t test and f test. Yes, we did them to determine the significance of the linear relationship that is being hypothesized.

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In order to do that, we had to make some assumptions regarding the error term epsilon in the regression model. These assumptions were critical in the hypothesis tests that we did. In both tests, at least for simple linear regression, the null hypothesis is that the coefficient beta 1 is equal to 0, and the alternate hypothesis is that beta 1 is not equal to 0.

Both hypothesis tests require estimates of the variance of epsilon, which we denoted as sigma squared. We use the data available to obtain an estimate of this sigma squared, which we call as s squared. And then, what did we do for the t test? For the t test, S square was computed as MSC. Yes, the MSC was defined as the SSE divided by its degrees of freedom, which was n minus 2.

The standard error for this estimate was the square root of the mean square error. The test statistic, a t random variable, was computed to be b1 divided by the standard error. Using this value of the t statistic, we computed the p value and we reject the null hypothesis if this p value is small enough.

Rejecting the null hypothesis allows us to claim that beta1 is not equal to 0, and therefore a linear relationship between x and y is significant. The f test was very similar. However, it used a different estimate of sigma squared. What was it Tejas? The sigma squared estimate was derived as the mean squared regression or MSR.

Yes, which was SSR divided by the degrees of freedom. And if the null hypothesis were true, the ratio of MSR over MSE would be equal to one. This test statistic was an F random variable. And if the null hypothesis were not true, then beta one would not be equal to zero. And the ratio of MSR over MSE would be greater than one.

We compute the test statistic, the F random variable and the associated P value. And if you reject the null hypothesis, we validate the significance of the linear relationship between X and Y. Finally, we looked at the analysis of the residuals to determine whether the assumptions that we made for the test of significance on the error epsilon were violated or not.

We looked at the plots of the residuals against the independent variable x. We also looked at plot of residuals against the dependent variable y. Yes, and we looked at the plot of the

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standardized value of the residuals to see if there were any outliers. So that was the theory. We also saw that all of these calculations can be done fairly quickly using the regression feature in the analysis tool pack in Microsoft Excel.

When we did that, what did we find? We found that the estimate of the regression parameters b0 and b1 were 1047. 37 and 85. 27 respectively. Yes, and therefore, the estimated regression equation is given by 1047. 37 plus 85. 27 times x. The model has an R square of 0. 65. That is, it explains 65 percent of the variation in y.

Is that a good percentage? I would think that it is a pretty good as a start. From these results, we also found that the correlation coefficient between the monthly spend y and the annual income x is the square root of 0. 65, which is 0. 806. What else did we find? We found that the standard error s is given by 868. 9. Yes, and the square of this quantity would be our estimate of sigma square. And what did the ANOVA table tell us? The F statistic computed as the ratio of MSR over MSE was 98. 25. Yes, and that implied a pretty low p value, which means that we could reject the null hypothesis that says that beta 1 is equal to 0.

Therefore, there is a relationship between x and y that is statistically significant. And finally, what did the t test tell us? To compute the test statistic, we divided b1 by the standard error, which was 85. 27 divided by 8. 6, which came to be 9. 91. This value of the t random variable implied a low p value, which means that we reject the null hypothesis that says that beta 1 is equal to 0, implying that a linear relationship between y and x is indeed statistically significant.

Why do two tests to decide the same thing? Well, Tejas, in the case of simple linear regression, it does appear to be redundant. But when we do multiple linear regression in the next module, you will see that the roles of the F test and the T test are a little bit different. So just wait. Okay. So what did we learn from our residual analysis?

We computed the standardized version of the residuals. Yes, and we found that most of these residuals were within plus 2 and minus 2. And there were two values that had standardized residuals that were slightly above 2 in an absolute sense. And what about the plots of the residuals against x and the plots of the residuals against y?

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Both did not indicate any concern in patterns. Yes, although they were not perfect bands, they did not have any funnel or curvilinear shape either. All right, excellent.

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SLR - Applications using Examples – I

Let us now look at another example. We will look at the same Hanumantha's data set and now try to do a linear regression of the monthly spent y against the household size x. Here, I have the data in the spreadsheet with the monthly spend in column D and the household size in column C. Let us first create a scatter plot of the household size on the horizontal axis and monthly spend on the vertical axis.

What do you notice, Tejas? I think there could be a linear trend here as well. Yes, I agree. So it might be worth considering a linear regression model. I agree. Let's plot a trend line. Yes, let's do that. And get the equation and the R square value as well. What equation do you get? The equation I get is that y is equal to 520.13 times x plus 2083. 7. Yes, and what is the R squared value? Oh, that is only 0. 39, much lower than what we got with household income. Yes, that is true. And what do you think that suggests? Is it saying that although there is a linear relationship, it is not as strong as the relationship between monthly spend and annual income?

Very likely, that is what it says. But before we jump into those conclusions, we need to go through the diagnostics. So let us use the analysis tool pack to do the regression analysis so that we can see the details. In analysis tool pack, we select y variable to be the data in column D, and the x variable to be the household size in column C.

In addition to pasting the output, let us also compute the residuals, the standardized residuals and the plot of the residuals. Did you get all of that? Just give me a minute. Okay, take your time. Okay, I got all the outputs. Now what? Now that we have the outputs, let us walk through them one by one. The R square value is 0. 396. That is the same value that we got through our scatter plot and trend line. This implies that our linear regression model is able to explain only 39 percent of the variation in the monthly spend as a function of the household size. In comparison, our model of monthly spent versus annual income did much better in explaining the variations in the monthly spent.

Now, what does the model tell us about the correlation between monthly spent and household size? We can just take the square root of the r square to find the correlation, right? Yes, Tejas. That's what we need to do. And what does that come out to be? It comes out to be equal to

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0.63. This correlation is much lower than the 80 percent correlation that we got between monthly spend and annual income.

That is right. And that's probably another reason why our linear regression model is not able to explain the variation in monthly spend as well as our earlier model. Next, let's look at the standard error. We see that the standard error is 1140. 71. The square of this quantity would be the estimate of the Sigma square for the error epsilon in the regression model.

Okay. Next, let us look at what the ANOVA analysis is telling us. What is it telling us? In the analysis we get the sum of square's regression and the sum of squares error. Yes. The sum of squares regression SSR is equal to 4 5 2 3 3 3 4 6. Since the degree of freedom for the mean square regression is equal to one.

The mean square regression or MSR is also equal to 4 5 2 3 3 3 4 6. What about the sum of squares error and the mean square error? The sum of squares error comes out to be 6 8 9 6 4 5 7 6, and the degree of freedom is 53 and minus two. That gives us a mean square error or MSE to be equal to 1 3 0 1 2 1 8. And so what does the F statistic come out to be?

The F statistic is the ratio of MSR over MSE. And that comes out to be 34.7. Yes. And what does that mean in terms of our P value? It's fairly close to zero. And so therefore we can conclude this allows us to reject the null hypothesis that beta 1 is equal to 0. Which implies that there is enough evidence to claim that beta 1 is not equal to 0.

Yes, and therefore, a linear relationship between monthly spend and household size is significant. Next, let us look at the table with the coefficients. What are b0 and b1? b0 is 2083. 7 and b1 is 520. 1. Yes, and these are the same values that we got in our scatter diagram. Now, let's look at the row corresponding to the household size and coefficient b1.

We notice that the standard error is 88. 2. And if we divide the coefficient b1, which is 520. 1, by the standard error of 88. 2, we get the t statistic, which is equal to 5. 89. For this t statistic, the p value is very small. And what does that imply? It implies that we can reject the null hypothesis again confirming that the linear relationship between x and y is significant.

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Let us now look at the residuals. We have the residuals as well as the predicted monthly spend given by our estimated regression equation. Notice that the predicted monthly spend is the average monthly spend for a given value of the household size. What can we say about these standardized residuals?

Here we can see that there are values of standardized residuals that are large. Yes, and in particular, there are three values that have standardized residuals that are greater than two. Does this mean that there are outliers? Yes, it looks like these are outliers. And we can see which points these are when we look at the scatter plot as well.

These points seem to be well above the regression line. And what does the plot of residuals against x tell us? The plot of residuals against x does not have a perfect band shape. But neither does it have a funnel or a curvilinear shape. Yes, that is right. And what about the plot of residuals against y? This plot is slightly different.

Yes, and although it is not a band or a funnel shape, it does seem to show a trend. Is that something to worry? Not something to worry, but it is telling us that there may be other factors that may be influencing the monthly spent. Such trends usually indicate that we need more independent variables in the linear regression model.

Yes, we know that household income impacts monthly spend too. Yes, and we will see their combined effect when we do multiple linear regression on the same data set in the next module. What about the plot of standardized residuals against x?

We see that it reveals the same issue that we noticed when we looked at the numbers. There are a few values of the standardized residuals that seem to lie above 2. Now, hopefully this gives you a good sense of how to use the regression analysis to understand the relationship between variables in the data.

Yes, this was definitely helpful and actually fun. It was nice to step away from the theory and just do the computation, make some interpretations and decisions. Yes, that is right. Most likely you're not going to worry about the theory every time you want to use regression. But once you

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understand the theory behind regression, you know how to interpret the outputs that a statistical software like Excel or any other will give you.

Shall we do some more examples? Sure. I think that would be helpful. Let's go and do another example.

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SLR - Applications using Examples – II

We will look at Buzzer Rogers Customer Satisfaction Survey Dataset. If you recall, we had looked at this dataset when we did hypothesis test in the last module. The dataset had 36 responses from a survey. It had variables like the score, the price of a computer, the brand, and features like RAM, hard disk, etc.

Basaraju is interested in knowing what factors might be influencing the satisfaction score by customers. He wants to know if the price is a significant determinant of this satisfaction score. So in this case, score would be our dependent variable, y, and price would be our independent variable, x. Let us now try and construct a linear regression of the satisfaction score y as a function of the price paid for the computer purchased x.

Here I have the data in the spreadsheet with the satisfaction score in column H and the price paid in thousands of rupees. in column G. Let us first create a scattered plot of the price paid on the horizontal axis and the satisfaction score on the vertical axis. And what do we observe? We see that there could be a linear trend.

So it might be worth considering a linear regression model. I agree. Let's plot the trend line. Yes, let us do that and get the equation and the R square value as well. What equation do you get? The equation I get is that y is equals to 48. 44 plus 0. 132 times x. Okay, and what is the R square value? That is only 0. 36, but we saw a good linear trend in the scatter plot. Yes, that is true. And how do you explain this low R square value? It is suggesting that there may be other factors than price that is causing the variation in these scores. Our regression analysis will also tell us whether the assumed relationship between the score and the price is significant.

So let us consider the regression analysis using the analysis toolpack and interpret the outputs. In the analysis tool pack, we select the Y variable to be the score data in column H and the X variable to be the price in column G. In addition to pasting the output, there is also compute the residuals, the standard residuals and the plots of the residuals.

Let me give you a minute to do all of that. Score in column H as Y and price in column G as X. Okay, got it. Yes. Now that we have these outputs, what do we look at first? Let us confirm

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the R square. I have 0. 36 from tool pack as well. Yes, and that is the same value that we got using our scatter plot and trend line.

So this implies that our linear regression model is able to explain only 36 percent of the variation in the score as a function of the price of the computer purchased. Satisfaction related to the purchase of computer could be a function of many other things other than the price paid. Now, what does the model tell us about the correlation between the score and the price?

We can just take the square root of the R square to find the correlation. Right? I get 0. 59. Yes, and this correlation is positive, not as high as we saw in Hanumantha's example, and that also perhaps explains our low value of r square that we get. Next, let's look at the standard error. We see that the standard error is 5.19. The square of this quantity would be the estimate of the sigma square for the error epsilon in the linear regression model.

Next, let us understand what the ANOVA output is telling us. The table shows that the sum of squares regression and the sum of squares error are 510.8 and 940.8 respectively. Yes, the sum of squares regression, SSR, is 510.8. And since the degree of freedom for the mean square regression is equal to 1, the mean square regression, or MSR, is also equal to 510. 8. What about the sum of squares error and the mean square error? The sum of square error is 914.8, and the degree of freedom is N minus two, which is 34, so that gives us a mean square error, or MSE of 26.9.

These numbers are much smaller than the one we got in Hanumantha's case. Yes, there we were looking at monthly spend which was in thousands and the square of those values. Here we are looking at scores which are between 0 to 100. Okay, now what does the F statistic come out to be? The F statistic is the ratio of MSR over MSE and that comes out to be 18. 9. Yes, and what does that mean in terms of our p value? It is close to zero, and therefore, we conclude that this allows us to reject the null hypothesis that beta 1 is equal to zero. Which implies that there is enough evidence to claim that beta 1 is not equal to zero. And therefore, a linear relationship between the score and the price is indeed significant.

Now, let's look at the table with the coefficients. What are B0 and B1? B0 is 48. 44 and B1 is 0. 132. Yes, and these are the same values that we got in our scatter diagram. And how do we

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interpret the coefficient B1? For every thousand rupee increase in price, the score increases by 0. 132, right? It is the slope with respect to the price, or as economists would say, B1 represents the marginal contribution.

Wait a minute, is this telling us that score increases as price increases? It should be the opposite. Lower the price, more the satisfaction. Tejas, that is a good observation. And that is our intuition, our understanding about human behavior. But regression is just reacting to the data in the training set.

So if the data is indicating that higher priced purchases led to higher satisfaction score, that is what the model will tell us. But is this coefficient significant? Is beta 1 equal to 0? What is the outcome of the hypothesis test with the null hypothesis of beta 1 equal to 0? Let's look at the row corresponding to the price and the coefficient b1 to answer these questions.

We notice that the standard error is 0. 03, and if we divide the coefficient b1, which is 0. 132, by the standard error 0. 03, we get the t statistic to be equal to 4. 36. And for this t statistic, what is the p value? It is very small. Yes. And what does that imply? It implies that we can reject the null hypothesis, again confirming the linear relationship between price x and the score y is indeed significant.

Let us now look at the residuals. We have the residuals as well as the predicted score given by our estimated regression equation. Notice that the predicted score is given by our estimated regression equation is the expected score for a given value of the price. And what can we say about the standardized residuals?

Here we can see that there are two values of standardized residuals that are less than minus 2. Yes, and in particular, these are the residuals with values of minus 2. 48 and minus 2. 12. In both cases, our model over predicts the score by quite a bit. Does this mean that there are outliers? Yes, it looks like there are outliers. We can see which points these are when we look at the scatter plots as well.

These points seem to lie well below our regression line. Now, what does the plot of residuals against x tell us? The plot of residuals against x does not have a perfect band shape. Yes, but

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neither does it have a funnel or a curvilinear shape. And what about the plot of residuals against y? Again, this plot is slightly different.

Similar to what we saw in the Hanumantha's example with household size. Yes, and although it is not a band or a funnel, it does seem to show a trend. It is telling us that there may be other factors that may be influencing the score other than the price. Such trends usually indicate that we need more independent variables in the linear regression model.

This is interesting. Feels like solving a mystery and being a detective. In a sense, that is what we are doing, learning and drawing inferences from the sample data about the behavior in the population. What about the plot of standardized residuals against x?

We see that it reveals the same issue we noticed when we looked at the numbers. There are a few values of standardized residuals that seem to lie below 2. Okay, that's about it then for this example. Hopefully that was helpful. Let us do another example. Where are we going next? We are going to meet Manjula Nayak, who works for 99 Acres.

If you recall, we had looked at Manjula Nayak's data set in Module 1. Here, you see an excerpt of the data set collected by Manjula Nayak, who works for 99 Acres. The data set corresponds to a household survey done in Bengaluru with around 500 observations. It has various types of information, including the number of members in the household, where they reside, their monthly income, monthly payments, and so on.

We will look at these fields in more detail next. The first column in this data has an identifier for each element. Here, it's just a number. It could very well have been the name of the head of the household. In the data set, we also have for each household variables of interest, like the family size, the location of their house, whether they are owning or renting a house, their income, their monthly payments, and their overall satisfaction in terms of where they are residing. Manjula is working with banks to help customers get home loans, and she wants to know what factors influence a household's debt. She wants to know if the household income is a significant determinant of the household debt.

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So what should be our dependent variables and independent variables? Debt should be our dependent variable, y? Yes, that is right. And what should be our independent variable? Debt can be influenced by several factors like income, ownership, monthly payments, utilities. Which one do we choose? Yes, we could choose several and we will do that in the next module on multiple linear regression.

For now, we need to pick one independent variable. So let us try first income. So in this case, first income would be our dependent variable x. Yes, now let us try to construct a linear regression of the debt y as a function of the first income x. Here we have this data in the spreadsheet with debt in column K and the first income in column G.

Shall we create a scatter plot first? Yes, always a good idea with the first income on the horizontal axis and the debt on the vertical axis. And what do we observe? We see that there could be a linear trend. I agree, but it is not as distinct as we saw in the other cases. For a given level of first income, there is a lot of variation in the debt.

Simple linear regression might struggle. Anyway, let's plot a trend line and let's get the equation and the r square value as well. What equation do we get? The equation I get is that Y is equal to 32976 plus 0. 04 times X. Yes, and what is the R square value? We get an R square value of 0. 31. It is not very high.

The R square value of 0. 31 indicates that the first income only explains about 31 percent of the variability observed in the household debt in the given dataset. Let us conduct the regression analysis using the analysis tool pack and interpret the outputs. In the analysis tool pack, we select the Y variable to be the household debt data in column K, and the X variable to be the first income in column G.

As we have done before, in addition to pasting the output, let us also compute the residuals, the standardized residuals, and the plots of the residuals. Let me know when you have done all of that.

Okay. Now that we have the outputs, what do we look at first? Let us look at the R square. We get a value of 0. 31 from tool pack. Yes, and this implies that our linear regression model is

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able to explain only 31 percent of the variation in the household debt as a function of the first income of the household.

Clearly, debt depends on other factors other than the first income, it appears. Now, what does the model tell us about the correlation between debt and first income? We can just take the square root of the r square and find the correlation, and you get 0. 56. This correlation is positive as we would expect. Next, let's look at the standard error.

We see that the standard error is 28, 619. The square of this quantity would be the estimate of the sigma squared for the error, epsilon, in the linear regression model. Let us now go to the ANOVA table's output. The table shows that the sum of squares regression and the sum of squares error, they're big numbers. Yes, SSR is 1. 86 times 10 raised to the power 11 and SSE is 4. 08 times 10 raised to the power 11.

One of the reasons they are big is because we have 500 data points. And since we are looking at simple linear regression with one independent variable, the degree of freedom for the mean square regression is equal to 1. And so the mean square regression, or MSR, will also be equal to 1. 87 times 10 raised to the power 11. Now what about the sum of squares error and the mean square error? The sum of squares error is 4. 08 times 10 raised to the power 11. And the degree of freedom will be equal to n minus 2. which in this case would be 498.

That gives us a mean square error or MSE of 819076096. And what does the F statistic come out to be? The F statistic is the ratio of MSR over MSE and that comes out to be 228. Yes, and what does that mean in terms of our P value? It is close to zero and therefore we conclude this allows us to reject the null hypothesis that beta1 is equal to zero.

Yes and that implies that there is enough evidence here to claim that beta1 is not equal to zero and therefore a relationship A linear relationship between debt and first income is significant. Now let's look at the table with the coefficients and what are the values that we get for B0 and B1. B0 is 32976 and B1 is 0.

04. Yes, and how do we interpret the coefficient B1? For every rupee increase in the first income, the debt increases by 0. 04. It's a slope with respect to the first income. As we saw

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earlier, B1 represents the marginal contribution. But is this coefficient significant? Is beta 1 equal to 0? And what is the outcome of the hypothesis test with a null hypothesis as beta 1 is equal to 0?

We need to look at the row corresponding to first income and its coefficient B1. Yes. And what is the standard error, the test statistic and the p value in this case? We notice that the standard error is 0. 002. And if we divide the coefficient b1, which is 0. 4, by the standard error of 0. 002, we get t statistic which is equal to 15.

1. Yes, and for this t statistic, what is the p value? It's very, very small. And what does that imply? It implies that we can reject the null hypothesis, again confirming that the linear relationship between first income and household debt, y, is significant. Let's now look at the residuals. We have the residuals as well as the predicted debt given by our estimated regression equation.

And what can we say about the standardized residuals? Here we can see that there are a few values of standardized residuals that are less than minus two and more than plus two.

Yes, by and large, they are within the range. And what does the plot of the residuals against x tell us? The plot of residuals against x looks like an oval. Yes, it does not have a perfect band shape, but neither does it have a funnel or a curvilinear shape. And what about the plot of residuals against y?

Again, this plot is slightly different, similar to what we saw in the Basavaraja's example of residuals versus score. Yes, and again, as it did in that example, it is telling us that there may be other factors that may be influencing debt other than the first income. Such trends, as we saw, usually indicate that we need more independent variables in the linear regression model.

We should come back to this example also in the next module. See how regression is helping us learn and infer from the sample data about the behavior in the population. Okay, what about the plot of standardized residuals against x?

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It reveals the same issue we noticed when we looked at the numbers themselves. There are a few values of standardized residuals that seem to lie above and below 2. Yes. Good. We learned something about household debt.

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SLR - Applications using Examples – III

Can we do a simple linear regression of debt versus another variable, just to compare? Sure. Which variable shall we pick? Monthly payments or utilities? Sure, but which one? Let us try monthly payment. The more the payments the households have, the higher their debt should be. Seems logical. Shall we try that? So, Ashwini and Tejas, what should our dependent variable and independent variables be in this case? We retain debt as our dependent variable y.

Yes, that is right. And the independent variable? And monthly payment is our independent variable x in this case. Yes. So now let's try and construct a linear regression of the debt y as a function of the monthly payment x. Here, I have this data in a spreadsheet. Debt is in column K and the monthly payment is in column I.

Oh, we forgot the scatter plot. Should we do that? I did it. The monthly payment on the horizontal axis and the debt on the vertical axis. And what do we observe? We see that it is very similar to what we got with first income. There could be a linear trend, but again, there is a lot of variation in the debt for a given level of monthly payment, very similar to our earlier regression effort.

And I would guess the output of our simple linear regression is likely to be very similar too. Anyway, let's plot the trend line and get the equation and the R squared value as well. And what equation do you get? The equation I get is that y is equal to 2 8 7 9 2 plus 2. 94 times x. Yes.

And what is the r square value? We get the r square value to be equal to 0. 37. It's not very high. And an r square value of 0. 37 indicates that monthly payment only explains about 37 percent of the variability involved or observed in the household debt in the given data set. If you recall, we got an r square of 0. 31 when we used first income. So this is better, only marginally better. Anyway, let's conduct the regression analysis using the analysis tool pack and interpret the outputs. In the analysis toolpack , we select the y variable to be the household debt in column K. And the x variable to be the monthly payment in column I.

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As we have done before, let's compute the residuals, the standardized residuals, and the plot of the residuals. Do you have all that Ashwini and Tejas? Okay, what do we look at first? Let us confirm the R square. We still get a value of 0. 37 from tool pack. Yes, and this implies our linear regression model is able to explain only 37 percent of the variation in household debt as a function of the monthly payments of a household.

Clearly, debt depends on factors other than the monthly payments and first income. Maybe there are other factors too. Now, what does the model tell us about the correlation between debt and monthly payment? We take the square root of the r square to find this correlation. And in doing so, we get a value of 0. 60. This correlation is also positive, just as Tejas had guessed. Now let's look at the standard error. We see that the standard error is 27518. The square of this quantity would be our estimate of sigma squared for the error epsilon in our linear regression model. Next, let's go to our ANOVA tables output.

The table shows that the sum of squares regression and the sum of squares error, again, they are big numbers. Yes, the SSR is 2. 18 times 10 raised to the power 11 and the SSE is 3. 8 times 10 raised to the power 11. We saw earlier too that they were big. And partly because we have 500 data points, and both x and y are in thousands.

Now, since we are looking at a simple linear regression with one independent variable, the degree of freedom for the mean square regression is equal to 1. So therefore, the mean square regression, or MSR, is also going to be equal to 2. 18 times 10 raised to the power 11. And what about the sum of squares error and the mean square error?

In this case, the sum of squares error is 3. 8 times 10 raised to the power 11. The degree of freedom here would be n minus 2, which would come out to be 498. So that gives us a mean square error, or MSE, of 75, 72, 68, 785. And what does that mean in terms of the F statistic? The F statistic is the ratio of MSR over MSE, and that comes out to be 288.

Yes, and what does that mean in terms of the P value? Here too it is close to zero. Yes and therefore we conclude that we can reject the null hypothesis that beta one is equal to zero and a linear relationship between debt and monthly payment could be significant. Now let's look at the table with the coefficients and what values do we get for b zero and b one.

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b zero is two eight seven nine one and b one is 2.94. Yes and how would we interpret the coefficient b one. For every rupee increase in a monthly payment, the debt increases by 2. 94. It is a slope with respect to the monthly payment. But you see, when there are several factors influencing Y and we have only one independent variable X, it is hard to say that this is the marginal effect of only that one independent variable.

But that is what we can say at this point, and we will do much better in the next module on multiple linear regression. Anyway, is b1 significant? Is beta1 equal to zero? What is the outcome of the hypothesis test with the null hypothesis that beta1 is equal to zero? We need to look at the row corresponding to monthly payment and its coefficient b1.

Yes, and what is the standard error, test statistic, and p value? We notice that the standard error is 0. 174, and if we divide the coefficient b1, which is 2. 94, by the standard error of 0. 174, we get the t statistic. And what is its value? The value of the t statistic is equal to 16. 95. Yes, and for this t statistic, what do we get as the p value?

It's pretty close to zero. That implies that we can reject the null hypothesis, again confirming the linear relationship between the monthly payment x and the household debt y. Let's now look at the residuals. We have the residuals as well as the predicted debt given by our estimated regression equation.

And what can we say about our standardized residuals? Here we can see that there are several values of standardized residuals that are less than minus 2 and more than plus 2. Yes, and what does the plot of the residuals against x tell us? The plot of residuals against x looks like an oval, very similar to what we got for the first income.

Yes, it does not have the perfect band shape that we want, but neither does it have the funnel or the curvilinear shape. And what about the plot of the residuals against y? Oh, this plot is similar to what we have been seeing, clearly a case for multiple linear regression, is it not? Yes, it is telling us that there may be other factors that may be influencing debt other than monthly payment.

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We will definitely come back to this example in the next module. I cannot wait to see what multiple linear regression tells us for this example. Yes, I agree. See how the regression model is helping us infer from the sample data. Now, what about the plot of the standardized residuals against x? It reveals the same issue that we noticed when we looked at the numbers themselves.

There are several values of standardized residuals that seem to lie above and below 2. Yes, good. We can always do another example on simple linear regression with that versus another variable. But I think we are getting ready for multiple linear regression. I agree. Okay, so let us summarize what we have learned and then look at multiple linear regression.

Before we close this module, let's summarize what we have learned in this module. We started this module looking at covariance and correlation. We talked about the expressions for correlation and covariance. We also talked about explainable correlation and spurious correlation. Then, we moved on with our journey of statistical inference by looking at simple linear regression.

In simple linear regression, a single dependent variable y is related to a single independent variable x. In simple linear regression, the regression model is y is equal to beta 0 plus beta 1 times x plus epsilon. The expected value of y, or E of y, is equal to beta 0 plus beta 1 times x. This equation describes how the expected value of y is related to x.

We then talked about the various steps involved in building a simple linear regression model. Then we use sample data and least squares method to develop the estimated regression equation y hat is equal to b 0 plus b 1 times x. Actually, B0 and B1 values obtained from linear regression are sample statistics that are used to estimate the unknown model parameters beta 0 and beta 1 in the population.

Then we wanted to validate our model. And to do that, we defined the concept of coefficient of determination as a measure of the goodness of fit of our model. The coefficient of determination is a ratio that tells us the proportion of the variation of the dependent variable y that can be explained by the estimated regression equation.

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We saw that in this process of validation, the assumptions behind the linear regression and especially the assumptions about the error term epsilon are critical for the T test and the F test used in the validation. These tests then told us whether the relationship between the two variables, Y and X are significant.

Then we used the analysis tool pack to conduct regression analysis on several examples using Excel. We saw how to analyze and interpret the results. We also saw how to conduct residual analysis to validate the assumptions, identify outliers, and influential observations. We have come far, have we not, Tejas and Ashwini?

Yes, it seems to be coming together. I liked how hypothesis tests, confidence intervals, and other things we learn about sampling comes together in a linear regression. Yes, they do. It also has probability elements. T distribution, F distribution, P values. Now I realize why we did all of that first before linear regression.

Yes, and we will see more interesting applications in the next module. In the next module, we will expand the simple linear regression concept to multiple linear regression. where the dependent variable y will depend on more than one independent variable x.

Thank you Tejas and Ashwini for all your help and support, and thank you all for your attention. I will see you next time.

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